

Ryan Cho

Senior Software Engineer | Backend, APIs & Data Systems

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PROFESSIONAL SUMMARY

Software engineer with 7+ years building backend services, APIs, and data systems in fintech and capital markets. Deep experience in TypeScript, Python, Java, SQL, and React across payment-sensitive, accuracy-critical domains. Proven ownership of data modeling, ETL pipelines, vendor integrations, and cloud infrastructure at scale. Strong collaborator with Product and compliance stakeholders; committed to reliability, type safety, and clear documentation.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected **TypeScript microservices and GraphQL APIs** powering portfolio analytics, investor reporting, and payment-adjacent fund operations for institutional clients across multiple product domains.
- Owned **PostgreSQL data models** and transformation pipelines aggregating **100M+ position and transaction records** with layered correctness checks at each ingestion and reconciliation stage.
- Integrated **external vendor SDKs** from custodians, payment processors, and market data providers using idempotent webhook handlers and automated exception queues for failed settlements.
- Built **React dashboards and internal tooling** for operations and portfolio managers, replacing manual spreadsheet workflows and shortening monthly close cycles for cross-functional teams.
- Drove **ETL reliability improvements** through schema versioning, backfill tooling, and data quality monitors that surfaced discrepancies before downstream reporting consumers were impacted.
- Raised engineering standards through **rigorous code review** and mentorship on test coverage, error handling, and maintainable abstractions in accuracy-sensitive production systems.
- Partnered with Product from **ideation through production release**, translating business goals into technical specifications with explicit edge cases, scale assumptions, and rollback plans.
- Enhanced **CI/CD pipelines and Terraform modules** on AWS, improving deployment consistency and reducing environment drift between staging and production infrastructure tiers.
- Authored **runbooks and API documentation** used by engineers and automation workflows, accelerating onboarding and reducing recurring support requests for platform services.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Built **Java and Python backend services** supporting trade reporting and reconciliation pipelines processing millions of daily transactions across global capital markets operations.
- Owned **relational data models and SQL ETL jobs** that consolidated ledger and position data from upstream vendor feeds into audit-ready staging layers for regulatory reporting.
- Developed **REST APIs** consumed by front-office and risk applications, enforcing type-safe contracts and validation rules to protect financial data accuracy at service boundaries.
- Implemented **batch and event-driven pipelines** that improved end-of-day reporting turnaround while maintaining strict reconciliation tolerances across counterparty and internal books.
- Partnered with compliance teams to embed **PII masking, access controls, and audit logging** into services handling sensitive client, counterparty, and transaction metadata.
- Led **code reviews and technical specifications** for cross-team initiatives spanning three engineering groups and multiple external vendor SDK integrations.
- Contributed to **CI/CD automation** with Jenkins and GitHub Actions, plus infrastructure templates that standardized provisioning across development, UAT, and production environments.

- Supported **production incident response** for accuracy-sensitive reconciliation systems, diagnosing root causes and deploying fixes with documented postmortems and preventive guardrails.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Developed **Node.js and React features** for a B2B hospitality platform, supporting real-time ticket and inventory updates across client venues during peak service periods.
- Integrated **third-party payment SDKs** and webhook handlers to synchronize transaction state with internal order records under tight latency and idempotency constraints.
- Wrote **unit and integration tests in TypeScript** that expanded regression coverage for checkout, refund, and fulfillment flows prior to each production release.
- Partnered with senior engineers on **SQL schema revisions** and migration scripts that preserved data integrity while rolling out pricing and catalog changes.
- Contributed to **React-based internal admin tooling** enabling operations teams to configure business rules without direct engineering intervention.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Designed and shipped **responsive web interfaces** for nonprofit programs, improving donor engagement workflows across mobile and desktop browsers.
- Built **RESTful endpoints and CMS integrations** that reduced manual content updates by enabling program staff to publish updates through self-service tooling.
- Implemented **accessibility and performance optimizations** including semantic HTML and lazy loading that improved Lighthouse scores and broadened audience reach.
- Collaborated with stakeholders to translate **fundraising requirements** into technical specifications delivered on a fixed nonprofit project timeline.
- Established **Git-based deployment workflows** and contributor documentation that improved handoff quality for volunteer engineering collaborators.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, HTML5, CSS3, Redux, GraphQL Client, Responsive UI, Accessibility (WCAG)

Backend: Node.js, TypeScript, Python, Java, REST APIs, GraphQL, Microservices, Event-Driven Architecture

Databases: PostgreSQL, SQL Server, MySQL, Redis, Relational Data Modeling, ETL Pipelines, dbt

Cloud & DevOps: AWS, GCP, Terraform, Docker, Kubernetes, GitHub Actions, CI/CD

Monitoring & Tools: Datadog, Grafana, Prometheus, ELK Stack, Splunk, Jest, Pytest

Workflow & PM: Agile/Scrum, Jira, Confluence, Technical Specs, Code Review, Git